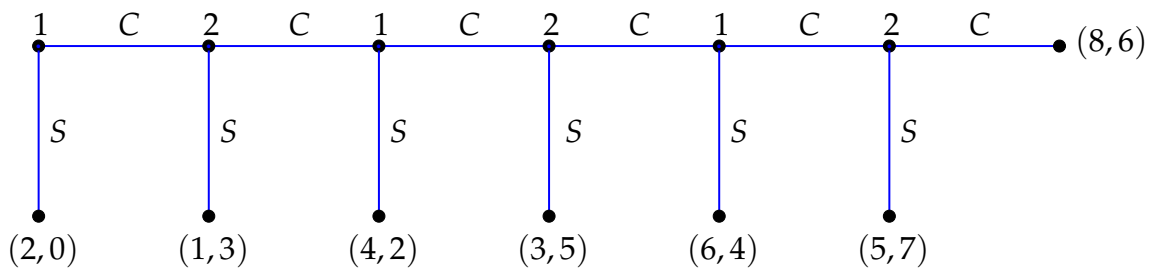
MTH636A (2025-26, Odd Semester)

Problem Set 3

1. In the centipede game, the two players move alternatingly. On each move, a player can stop (S) or continue (C). On any move, a player is better off stopping the game than continuing if the other player stops immediately afterward, but is worse off stopping than continuing if the other player continues, regardless of the subsequent actions. The game ends after a finite number of periods. Consider an example of this game below.



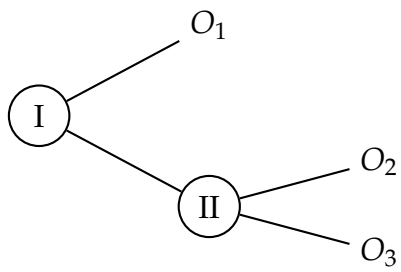
   (a) Determine the backward induction or subgame perfect equilibrium of this game. What is the associated outcome?

   (b) Show that there are other Nash equilibria, but that these always result in the same outcome as the subgame perfect equilibrium.

2. Depict the following situation as a game in extensive form. Peter and his three children, Andrew, James, and John, manage a communications corporation. Andrew is the eldest child, James the second-born, and John the youngest of the children. Two candidates have submitted offers for the position of corporate accountant at the communications corporation. The choice of a new accountant is conducted as follows: Peter first chooses two of his three children. The two selected children conduct a meeting to discuss the strengths and weaknesses of each of the two candidates. The elder of the two children then proposes a

candidate. The younger of the two children expresses either agreement or disagreement with the proposed candidate. A candidate is accepted to the position only if two children support his candidacy. If neither candidate enjoys the support of two children, both candidates are rejected.

3. Consider the game in the following figure



where O_1, O_2, and O_3 are distinct and taken from the set $\{(1,-1),(-1,1),(0,0)\}$.

   (a) Is there a choice of O_1, O_2, and O_3 such that Player I can guarantee victory for himself? Justify your answer.

   (b) Is there a choice of O_1, O_2, and O_3 such that Player II can guarantee victory for himself? Justify your answer.

   (c) Is there a choice of O_1, O_2, and O_3 such that both players can guarantee for themselves at least a draw? Justify your answer.

4. Recall the CHOMP game by David Gale. Answer the following questions.

   (a) Which of the two players has a winning strategy in a game of Chomp played on a $2 \times \infty$ board? Justify your answer. Describe the winning strategy. (Recall that the proof we did in class holds under the assumption that the board is finite.)

   (b) Which of the two players has a winning strategy in a game of Chomp played on an $m \times \infty$ board, where m is any finite integer? Justify your answer. Describe the winning strategy.

5. Consider the following game with perfect information and without chance moves, where for each of the subgames the alternative that holds (from von Neumann's theorem) is written within the subgame (see Figure 1). Find which alternative according to von Neumann's theorem is true for this game. Also, find the strategy/strategies?

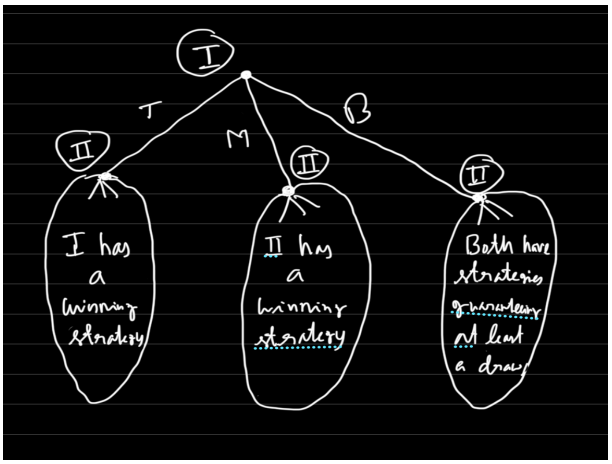


Figure 1

6. Consider the following game (known as the Nim game in the literature). It is a game between two-player where two piles of matches are placed before the players, and each pile has three matches. Each player in turn chooses a pile and removes any number of matches from the pile he has selected (he must remove at least one match). The player who removes the last match wins the game. Note that by von Neumann's Theorem (Theorem 1 in the slides), one of the players has a winning strategy in this game.

   (a) Find who has a winning strategy and find the winning strategy.

   (b) What will happen in the general case, i.e., when the two plies have x and y number of matches for some $x, y \in \mathbb{N}$.

7. Allen and Rick need to divide \$100 between them as follows: first Allen suggests an integer x between 0 and 100 (which is the amount of money he wants for himself). Rick, on hearing the suggested amount, decides whether to accept or reject. If Rick accepts, the payoff of the game is $(x, 1-x)$: Allen receives x dollars, and Rick receives $100 - x$ dollars. If Rick chooses to reject, neither player receives any money.

   (i) Describe this situation as an extensive-form game.

   (ii) What is the set of pure strategies each player has?

   (iii) Show that any result $(a, 100-a)$, $a \in \{0, 1, \ldots, 100\}$, is a Nash equilibrium payoff. What are the corresponding equilibrium strategies?

   (iv) Find all the subgame perfect equilibria (in pure strategies) of this game.

8. Establish whether there exists a two-player game in extensive form with perfect information, and possible outcomes I (Player I wins), II (Player II wins), and D (a draw), whose strategic-form description is

		Player II			
		s_{II}^1	s_{II}^2	s_{II}^3	s_{II}^4
Player I	s_{I}^1	D	I	II	I
	s_{I}^2	I	II	I	D
	s_{I}^3	I	I	II	II

If the answer is yes, describe the game. If not, explain why not.

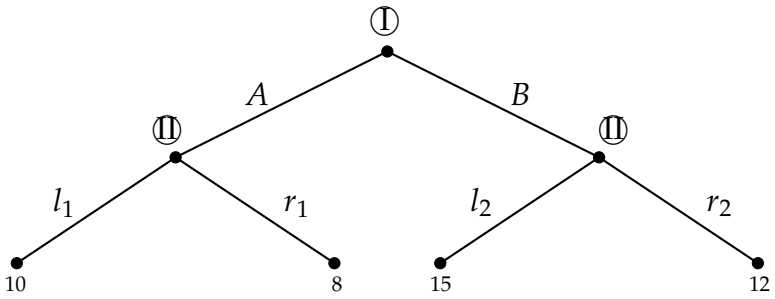
9. A pack of three cards, labelled 1, 2, and 3, is shuffled. Sachin, Sourav, and Rahul each take a card from the pack. Each of the two players holding a card with low values (1 or 2) pays the amount of money appearing on the card he or she is holding to the player holding the high-valued card (namely, 3). Depict this situation as a game in extensive form.

10. Establish whether there exists a two-player game in extensive form with perfect information, and possible outcomes I (Player I wins), II (Player II wins), and D (a draw), whose strategic-form description is

		Player II			
		s_{II}^1	s_{II}^2	s_{II}^3	s_{II}^4
Player I	s_{I}^1	D	D	D	D
	s_{I}^2	I	I	II	II
	s_{I}^3	II	II	D	D

If the answer is yes, describe the game. If not, explain why not.

11. Consider the following two-player zero-sum game in extensive form game:



(a) Find an optimal strategy for each player by applying backward induction.

(b) Describe this game in strategic form.

(c) Find all the optimal strategies of the two players.

(d) Explain why there are optimal strategies in addition to the one you identified by backward induction.